

DATA STORYTELLING

STATISTICAL VALIDATION

Data Analytics Internship

Apex Planet Software Pvt.Ltd

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EXECUTIVE SUMMARY



INFRASTRUCTURE

Successfully unified 9,994 transaction records into a **MySQL** pipeline, ensuring data integrity.



INTELLIGENCE

Identified core operational KPIs and regional bottlenecks via **Power BI** interactive modeling.



VALIDATION

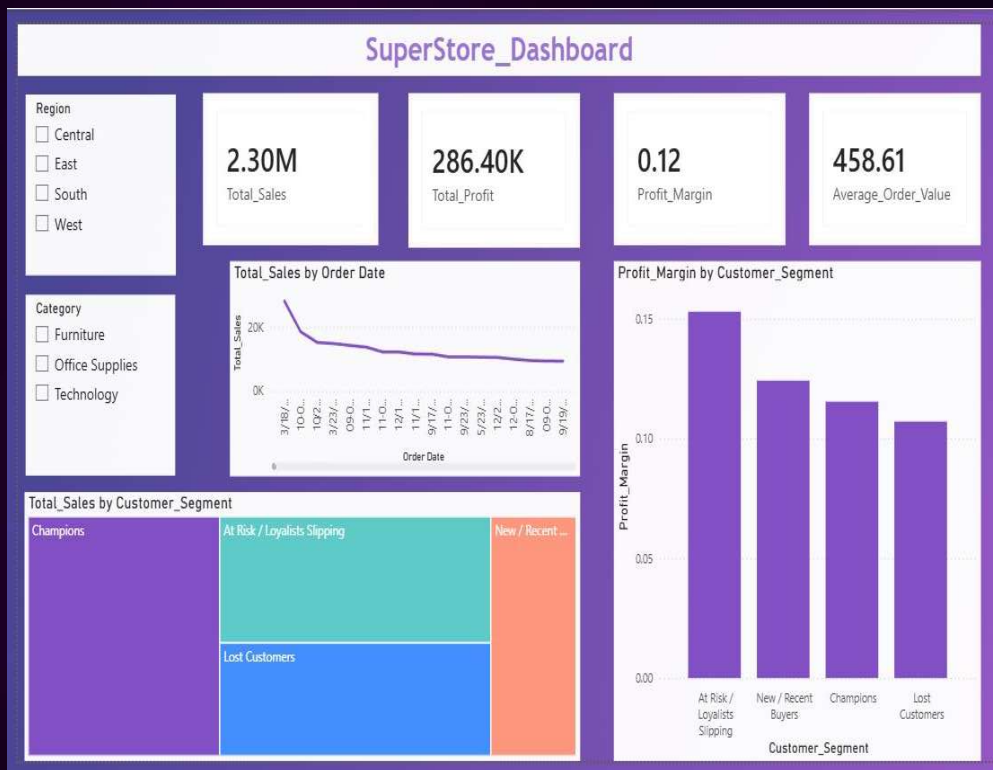
Used **T-Tests** to mathematically prove purchasing behaviors, removing business guesswork.

DATA PIPELINE ARCHITECTURE

- ✓ **Extraction:** Raw records migrated to centralized MySQL schema.
- ✓ **Cleaning:** Handled missing values and standardized data via Pandas.
- ✓ **Mockups:** Designed low-fidelity Excel wireframes for KPI prioritization.



OPERATIONAL KPI MONITORING



KEY DISCOVERIES

Our star-schema model in **Power BI** revealed:

Profit margins fluctuate significantly by product sub-category.

Regional shipping delays are the primary driver of customer churn.

THE SCIENTIFIC HYPOTHESIS

"Do Corporate clients spend significantly more per order than regular Consumers?"

NULL (H_0)

Mean Sales are identical across segments
($\mu_1 = \mu_2$).

ALTERNATIVE (H_A)

Mean Sales differ significantly ($\mu_1 \neq \mu_2$).

VALIDATION RESULTS (WELCH'S T-TEST)

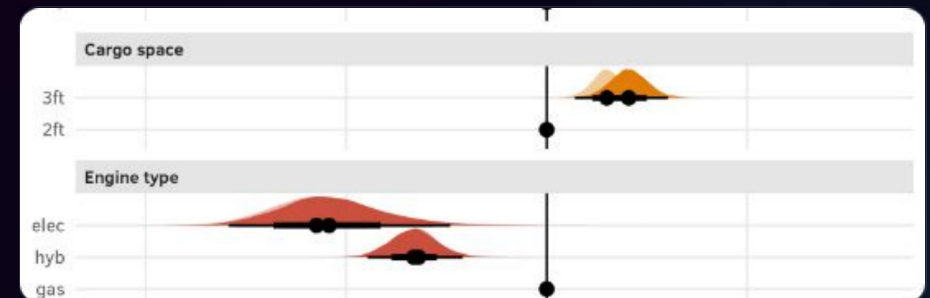
0.458

P-VALUE ($P > 0.05$)

THE MATHEMATICAL VERDICT

Across **8,211 combined orders**, we failed to reject the Null Hypothesis.

Conclusion: Purchasing power per cart is uniform. Observed gaps are purely random sampling variance.



STRATEGIC CALL TO ACTION



STOP BUDGET LEAK

Cancel plans for custom pricing tiers for Corporate accounts; their cart value matches Consumers.



PIVOT TO FREQUENCY

Redirect budget to **Subscription Models** to increase order volume rather than size.



FIX LOGISTICS

Tighten SLAs in high-churn regions identified in Task 3 to protect existing retention.

Thank you

Apex Planet Software Pvt.Ltd

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[https://github.com/KancharlaNithyasri/Apex-Planet-Data-A
nalyst-Task-4](https://github.com/KancharlaNithyasri/Apex-Planet-Data-Analyst-Task-4)